

Magnetic Levitation through Superconductors

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Abstract

Superconductors are materials, which exhibit zero resistance and expel magnetic fields at extremely low temperatures. The type of superconductor we were researching was YBa₂Cu₃O₇ (Y123), which has superconducting properties at a temperature of 93K [Fig.1]. We used liquid nitrogen to cool our pellets down to these temperatures to test the levitation of the magnets [Fig.2]. This levitation is caused by the Meissner effect, which is when magnetic fields are expelled from the inside because surface currents flow without resistance. This creates magnetization in the superconductor, which is equal and opposite of the magnetic field from the magnet. This cancels the magnetic fields everywhere causing the superconductor to show perfect diamagnetism [Fig.3]. Our goal was simple: create a superconductor that can levitate a magnet at least 3mm.

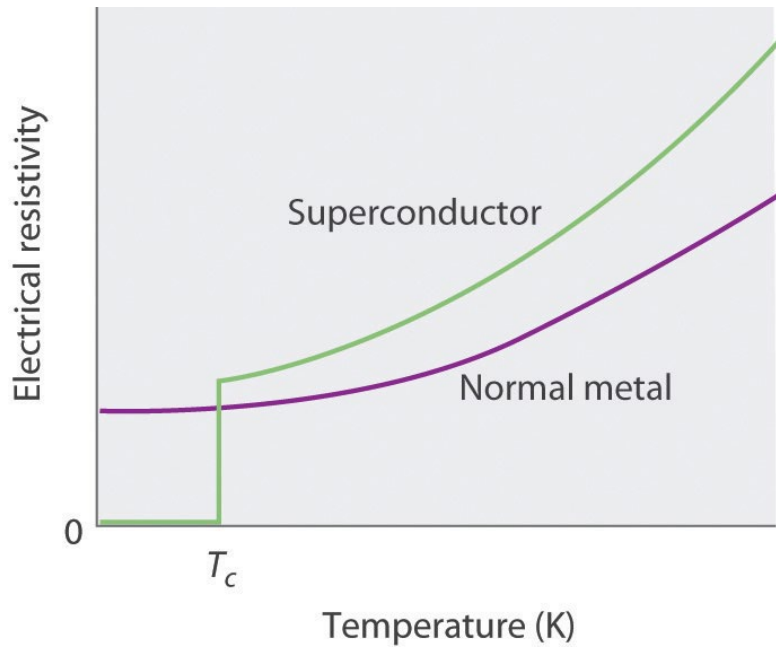


Figure 1: Electrical resistance of the material at different temperatures. [1]

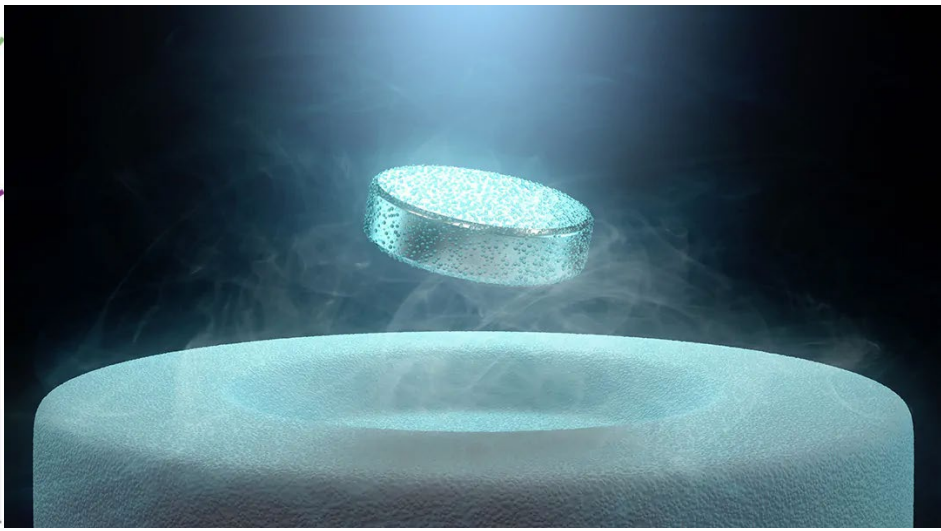


Figure 2: Example of magnetic levitation with a superconductor. [2]

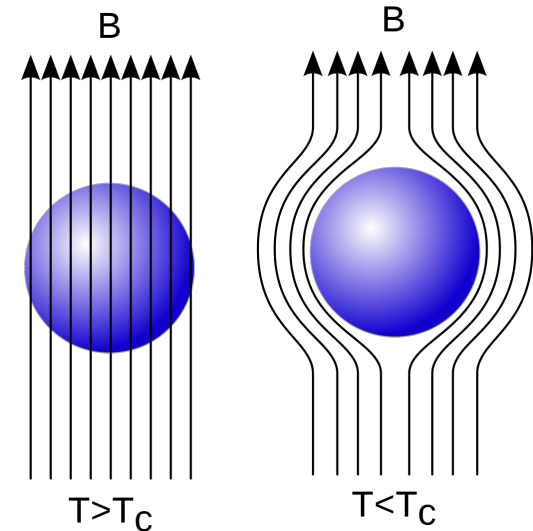


Figure 3: Demonstration of the magnetic field lines with the Meissner effect. [3]

Crystal Structure/Background

The final crystal structure of Y123 is an Orthorhombic structure, which means all the angles are 90 degrees and the edge lengths are all different (a by b by c). Thus, the structure is a rectangular prism. Before the material reaches this final crystal structure, the powders create a semi-conductor, which is a tetragonal structure. This structure is like the Orthorhombic except the base sides are the same length (a by a by c). There is a stacking sequence in the crystal along the c-axis: CuO–BaO–CuO₂–Y–CuO₂–BaO. These layers are separated by planes of CuO₂ with yttrium atoms between the copper-oxygen planes. It is within the copper-oxygen planes that superconductivity originates.

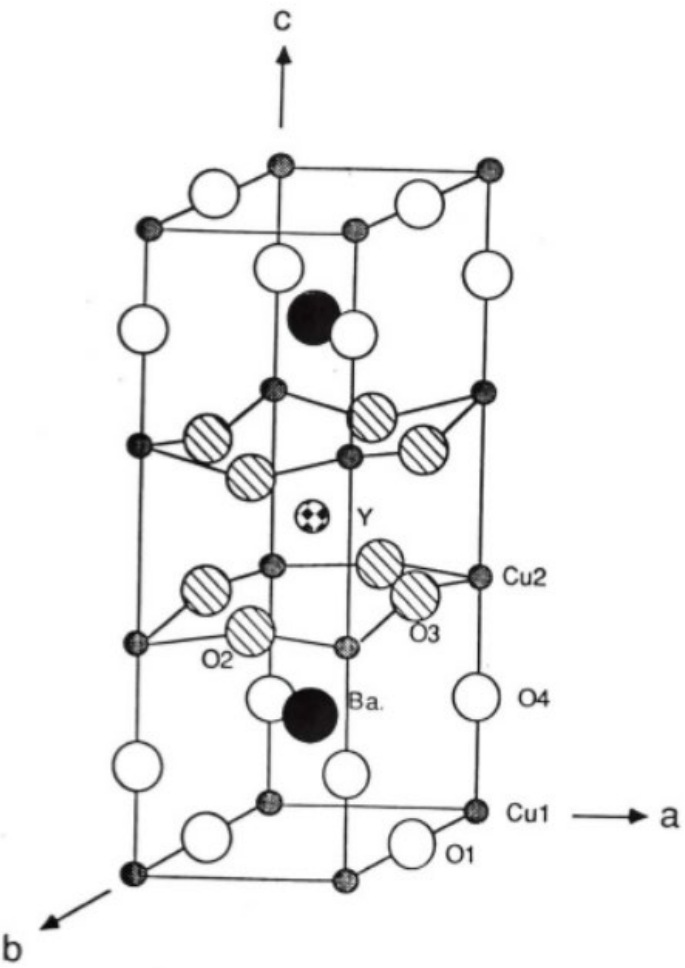


Figure 4: Crystal Structure of Y123

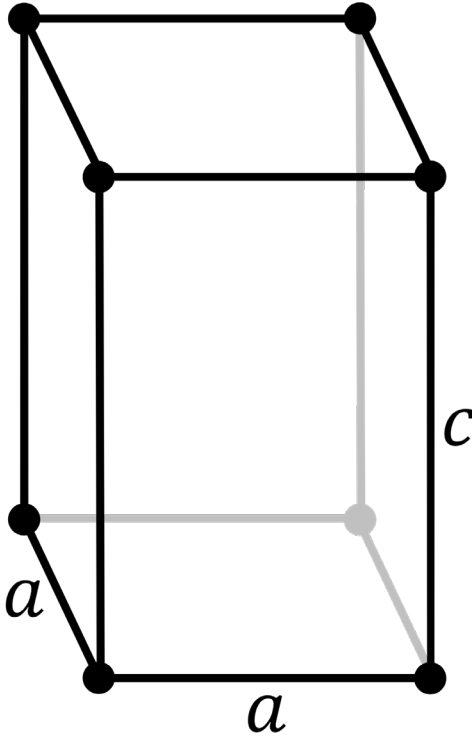


Figure 5: Tetragonal crystal structure

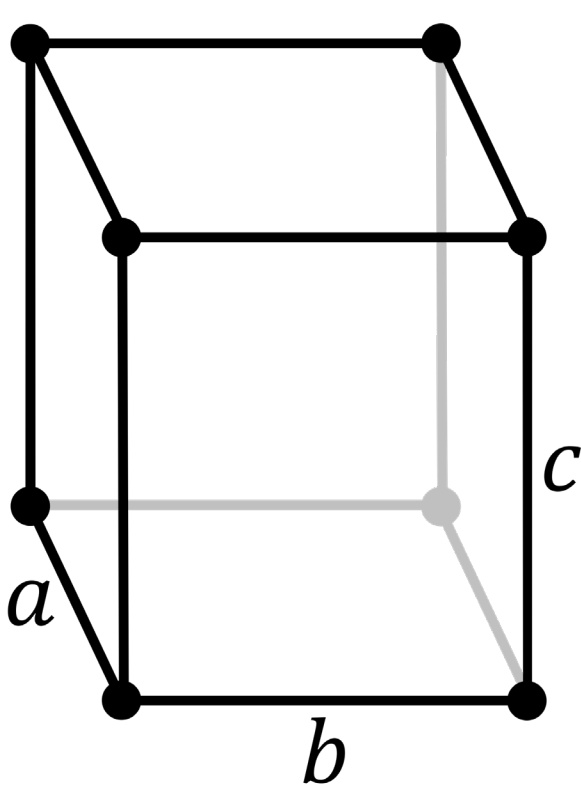


Figure 6: Orthorhombic crystal structure

Process

There are various methods for making the YBa₂Cu₃O₇, commonly known as the 123 superconductor. The two most common methods include the solid-state reaction and the coprecipitation method. In the solid-state method, oxide/carbonate forms of constituent materials are mixed, ground, and heated multiple times, while in the coprecipitation method, nitrate forms of the constituent materials are mixed, precipitated out of solution in their carbonate forms and is then heated. In this experiment the solid-state, also known as the "Shake and Bake" method will be used.

Solid-State Reaction Method:

1. Weigh the three constituent oxides/carbonate in the following quantities. Note the weights are based on the mole ratio of Y, Ba, and Cu needed.
 - a. Y₂O₃: 0.7565 g
 - b. BaCO₃: 2.6445 g
 - c. CuO: 1.599 g
2. Mix well using a ball mill. Shake for 10 minutes at a frequency of 30 1/s. Once complete open under fume hood and perform an initial XRD analysis on sample.
3. Transfer contents to a porcelain dish and place in a box furnace for 24 hours at 950 C.
4. Finely crush into a grind powder using the ball mill for 10 minutes at a frequency of 30 1/s. After completion, perform another XRD analysis.
5. After XRD confirms signs of YBa₂Cu₃O₇, the powder is pelletized. Using a 13 mm die, press the powder into a disc using 20 KN of force.
6. Place in tube furnace at 975 C for 18 hours with a steady flow of oxygen. After 18 hours, the pellet is cooled slowly at a rate of 1 C/min.
7. Test pellet in an insulated cup with liquid nitrogen and place a magnet to test for superconductivity.

Product

Each step in the shake and bake process is important. The first step is to after combining all the relevant starting powders is an initial firing, also called the calcination. During the beginning of the heat treatment, the barium carbonate is converted to barium oxide, releasing carbon dioxide in the process. The three oxides then crystallizes to form YBa₂Cu₃O_{6.5}. The result is a dark-grey powder. However, there are two other phases that may form including Y₂BaCuO₅, which is a green insulating compound, and BaCuO₂, another insulating compound. The second heat treatment after being pelletized is known as sintering. The pellet is heated and cool slowly in the presence of oxygen to convert YBa₂Cu₃O_{6.5} to YBa₂Cu₃O₇, at which point the crystal structure changes to become a superconductor.



Figure 7: Final product



Figure 8: Testing of final product for superconductivity in liquid nitrogen

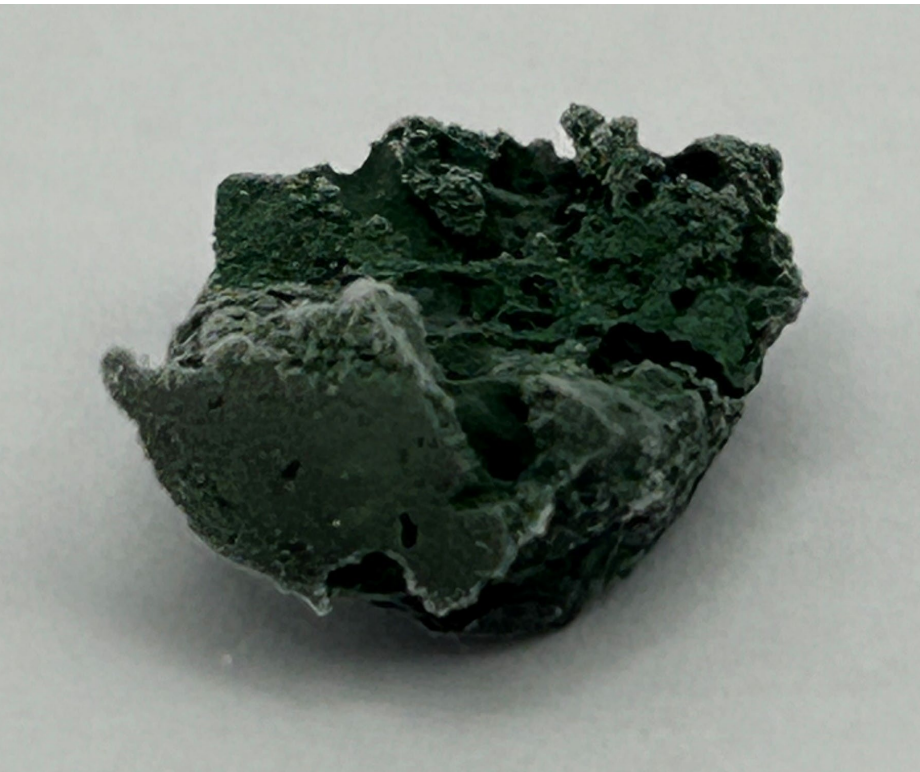


Figure 9: Failed attempt during sintering. Presence of insulating phases

Results

The X-ray Diffraction (XRD) Data for the raw components, after the first reaction, and after sintering.

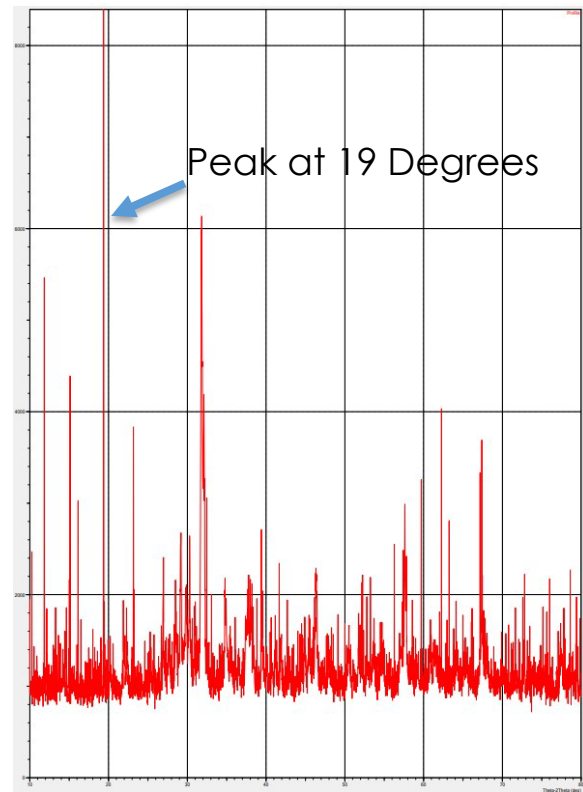


Figure 10: Mixture of Component Chemicals

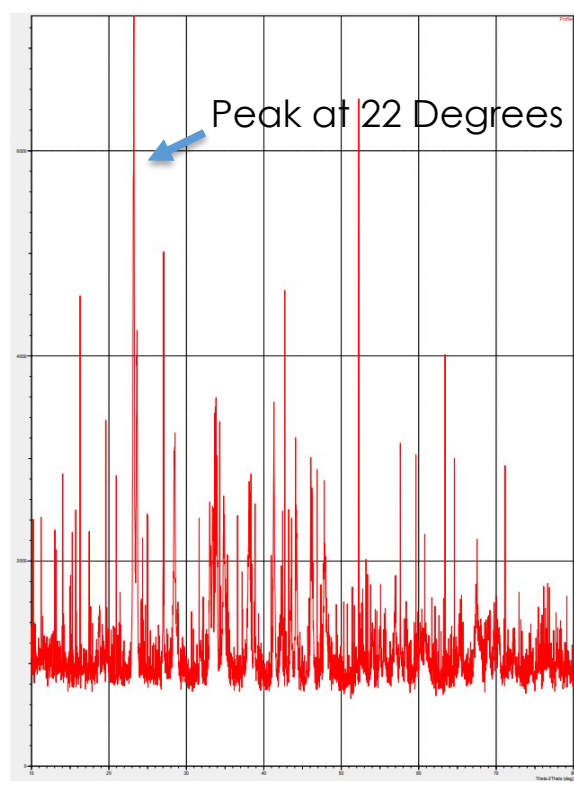


Figure 11: After Reaction

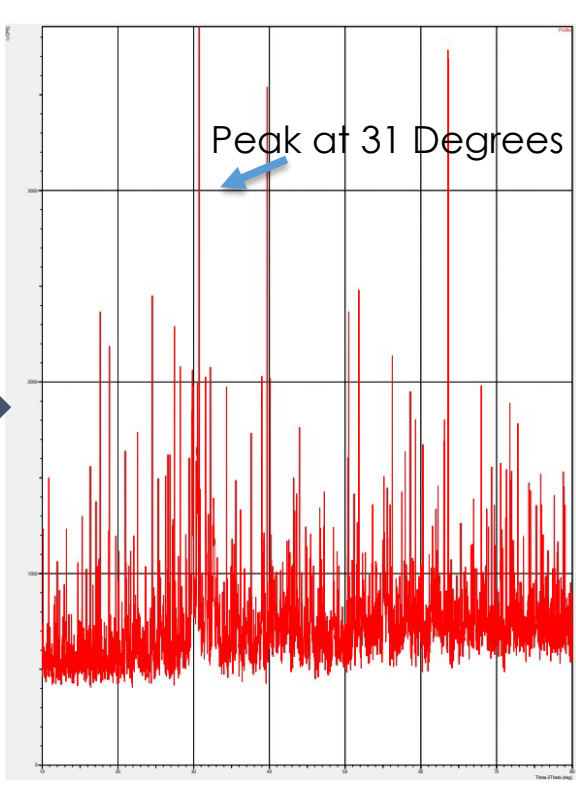


Figure 12: After Sintering

XRD scans show the changes in molecular and crystalline structures taking place in our compound after different processes. Pure Y123 has a peak at 22 Degrees which roughly matches the peak our XRD data shows after the reaction step.

Discussion

After running multiple batches of compounds and attempting to make many superconducting pellets, we were unable to make a superconducting pellet that would lift a small neodymium magnet. The three possible ways that could have lead to our supposedly superconducting pellets not superconducting are:

- Our process for creating the superconductor was flawed. While this is possible, we thoroughly researched how to create superconductors and our final process was the synthesis of that research. To rule out this possibility we could do further research into the methods of making Y123.
- The equipment and techniques we used to execute our process were incorrect or not precise enough. One possible example of this was that during the sintering process, some of our pellets melted which should not happen if the melting point of the pellet was above the set temperature of the furnace. This could mean that the temperature in the furnace where our pellets were was not at the set temperature. To determine if the equipment and techniques were the cause we could try running our process on different furnaces and determining if we get different results.
- How we determine if a pellet is superconducting by placing a magnet on top of the pellet and watching for levitation may not be sensitive enough to find low levels of superconductance. To address this possible issue, we could buy a magnetometer sensitive enough to pick up on the magnetic anomalies caused by the superconductor.

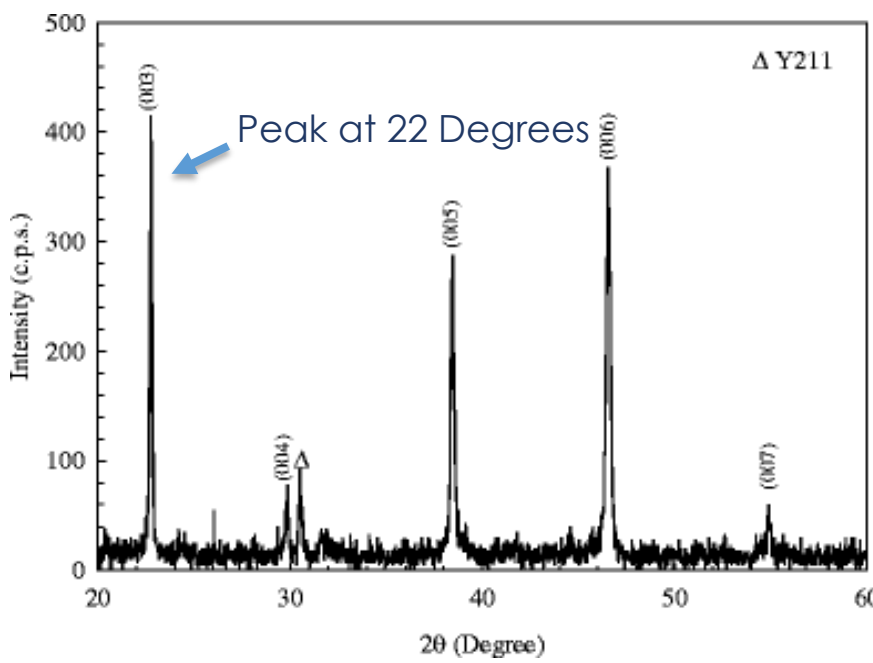


Figure 13: Expected XRD data for Y123 [4]

References

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- [4] *Fluctuation-Induced Conductivity Analysis of Y2O3-Layered YBCO Single ...* https://www.researchgate.net/publication/303906589_Fluctuation-Induced_Conductivity_Analysis_of_Y2O3-Layered_YBCO_Single_Crystal.